

REVATHY ISWARYA SEKARAN

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DATA SCIENTIST — DATA ANALYST — MACHINE LEARNING ENGINEER

Highlights

- Senior Quality Assurance professional with 12+ years of experience delivering and validating large-scale Web, Mobile, and API-driven enterprise systems across retail, healthcare, and media, with deep expertise in backend data validation, system integration testing, and ensuring data accuracy and integrity in production environments
- Recently completed M.S. in Data Science with rigorous training in statistical analysis, machine learning, predictive modeling, and applied analytics across structured, text, image, and time-series datasets.
- Hands-on experience in model development, evaluation, and deployment across supervised and unsupervised learning, deep learning, computer vision, and time-series forecasting, with additional experience applying transformer-based NLP and Large Language Models through prompting, fine-tuning, and evaluation workflows.
- Strong experience collaborating across business, engineering, QA, and analytics teams, with a focus on data quality assurance, root-cause analysis, Agile delivery, and translating complex technical and data findings into clear, actionable insights for stakeholders.

Core Skills

Languages & Big Data: Python (Pandas, NumPy, Scikit-learn), SQL, R, PySpark, Java, REST APIs

Machine Learning & Stats: Regression, Classification, Clustering, Ensemble Methods, Time Series, Hypothesis Testing, Bayesian Methods, Hyperparameter Tuning

Deep Learning & Computer Vision: PyTorch, TensorFlow, Keras, CNN, LSTM, YOLOv5/v8, Mask R-CNN, DeepSORT, TCGCN, FER, FaceNet, OpenCV

Natural Language Processing & LLM: Transformers (BERT, DistilBERT, MiniLM, FLAN-T5), Hugging Face, Large Language Models (LLMs), Prompt Engineering (Zero-shot, Few-shot), Instruction Tuning, LoRA/PEFT Fine-Tuning, Text Summarization, Text Preprocessing, Model Evaluation, Topic Modeling (LDA)

Visualization & Tools: Tableau, Power BI, Plotly, Streamlit, Git, JIRA, Agile, Advanced Excel (Pivot Tables)

Cloud & Platforms: AWS SageMaker (model training and experimentation on managed compute)

Software Engineering & QA: Web and Mobile Testing (Real Devices, Perfecto), Functional Testing, SIT, Regression Testing, UAT, Performance Testing, Mainframe Testing (JCL, DB2), SQL Backend Validation, API Testing (Postman, REST, SOAP), Automation Testing (Selenium WebDriver, TestNG, Cucumber)

Professional Experience

Research Assistant

Mar 2026 – Present

STRIDE Lab, University of Michigan–Dearborn

Michigan, USA

- Reviewed literature on JEPA architectures (I-JEPA, V-JEPA, LLM-JEPA) for self-supervised representation learning.
- Designed and evaluated experiments to assess JEPA models across domains, focusing on generalization and performance.
- Analyzed simulation outputs to identify causal relationships and support causal inference tasks.
- Developed evaluation metrics and documented research findings, presenting updates in weekly lab discussions.

Assistant Consultant (Technical Lead)

Jan 2012 – Sep 2024

Tata Consultancy Services (TCS)

Chennai, India

- Led end-to-end quality engineering and release validation across retail, healthcare, and media domains; mentored a 10-member team.
- Reduced defect leakage by 25% by identifying requirement gaps and improving early-stage validation.
- Performed functional and compatibility testing for complex web and mobile applications, including iOS/Android using real-device platforms (Perfecto).
- Executed complex SQL queries (joins, subqueries, aggregations) to validate backend data across workflows, APIs, and downstream systems.

- Led API testing and backend validation using Postman and Charles Proxy, ensuring system integration and data consistency.
- Built automation frameworks using Selenium WebDriver and TestNG, improving regression coverage and reducing manual effort.
- Collaborated with cross-functional teams in Agile environments, reporting quality metrics, risks, and defect trends.

Key Clients: Kaiser Permanente, Sirius XM, Dick's Sporting Goods, Target, The Home Depot

Projects

Harmonizing Machine Learning & Vocal Features for Parkinson's Disease Forecasting GitHub
Python, scikit-learn, TensorFlow, Streamlit, LIME

- Engineered acoustic features from 5,800+ voice samples and trained ElasticNet, SVM, Random Forest, ANN, and LSTM models with scaling and hyperparameter tuning to predict UPDRS scores.
- Best LSTM achieved $MSE = 0.002$ and $R^2 = 0.96$; deployed a Streamlit app with LIME-based explainability.

Student Engagement Prediction Using Facial Expression Recognition GitHub
Python, PyTorch, TensorFlow, FER, FaceNet, CNN, ShuffleNet, OpenCV

- Processed 9,000+ video snippets and extracted facial expression and embedding features using FER and FaceNet to classify student engagement.
- Trained CNN and ShuffleNet models with data augmentation, achieving 86% training accuracy.

Topic Modeling and Sentiment Analysis of Mental Health Conversations GitHub
Hugging Face Transformers, BERT, DistilBERT, MiniLM, LDA, NLP

- Applied LDA topic modeling and fine-tuned transformer models (DistilBERT with class-weighted loss and MiniLM with SMOTE) to classify mental health conditions from therapist Q&A and social media text.
- Achieved **ROC AUC up to 0.99** for Anxiety, Normal, and Personality Disorder classes using multi-class evaluation.

Spatio-Temporal Traffic Flow Prediction Using Surveillance Cameras GitHub
Python, Requests, BeautifulSoup, YOLOv5, Mask R-CNN, DeepSORT, TCGCN, PyTorch, OpenCV

- Collected traffic camera data via web scraping and built an end-to-end pipeline for vehicle detection and tracking using YOLOv5, Mask R-CNN, and DeepSORT.
- Extracted traffic features and applied a Temporal Convolution Graph Convolutional Network (TCGCN) to forecast traffic flow ($MAE = 6.53$ clear vs 9.12 rain).

Education

University of Michigan–Dearborn — M.S. in Data Science Aug 2024 – Dec 2025
Michigan, USA GPA: 3.95 / 4.0

VIT Bangalore — Post Graduate Program in Data Science 2023 – 2024
Bengaluru, India GPA: 4.0 / 4.0

Kalasalingam University — B.Tech in Computer Science & Engineering 2007 – 2011
Tamil Nadu, India CGPA: 7.4 / 10

Awards and Recognition

On-the-Spot Award — Star of the Quarter — Special Initiative Award — Best Team Award

Certifications and Courses

Scrum Master — ISTQB Foundation Level — ISTQB Advanced Test Analyst
 Databases and SQL for Data Science with Python — IBM (Coursera)